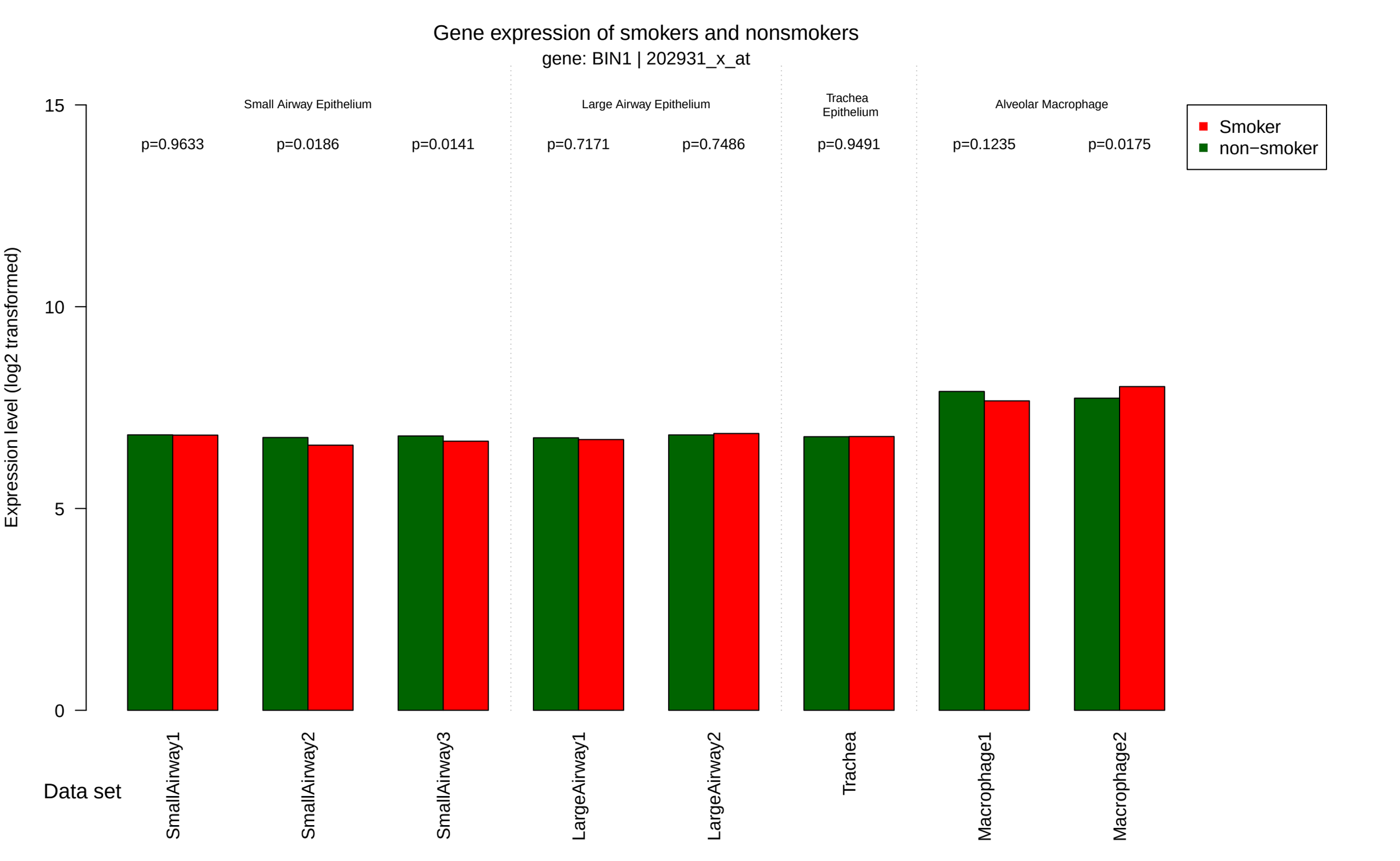
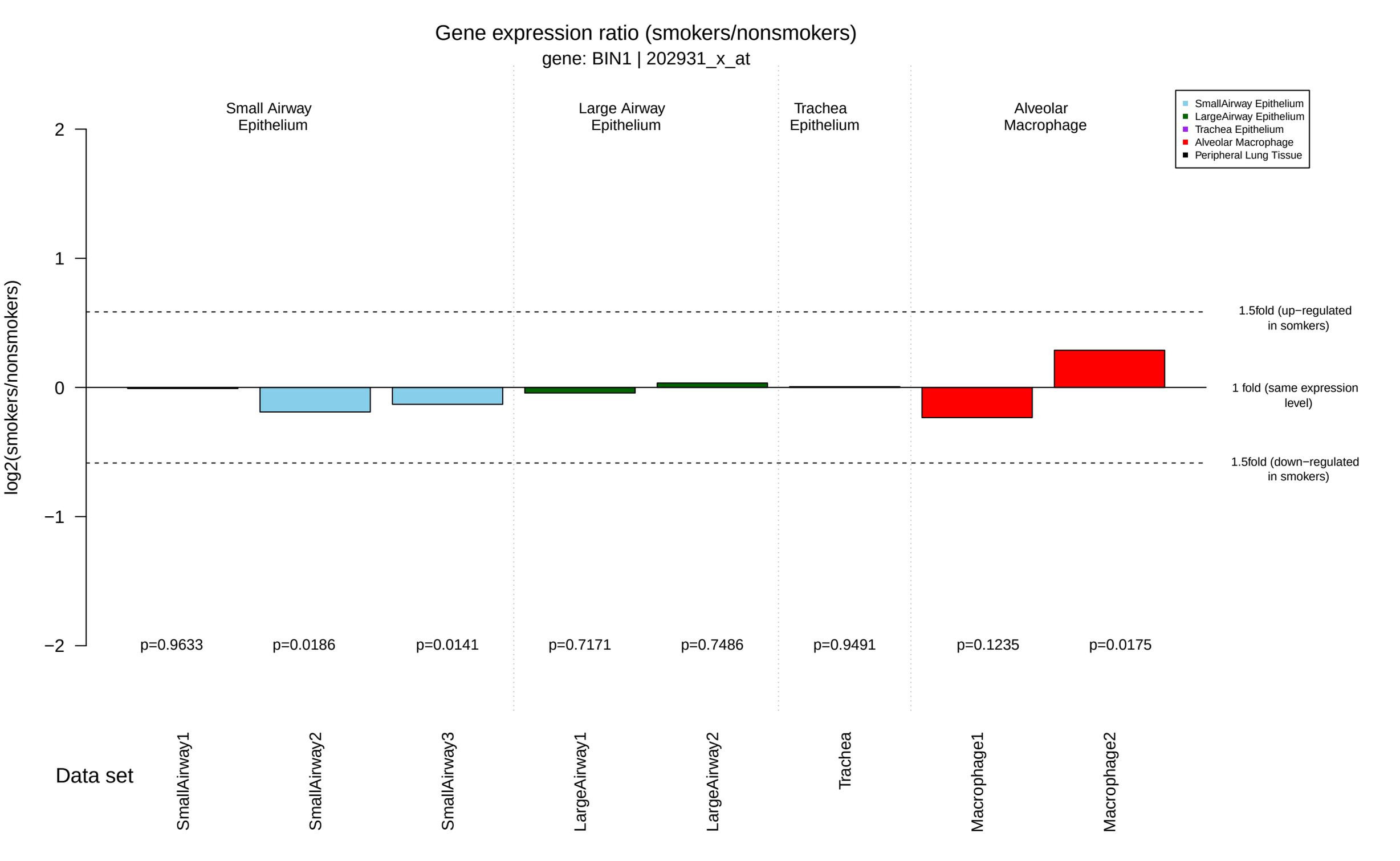
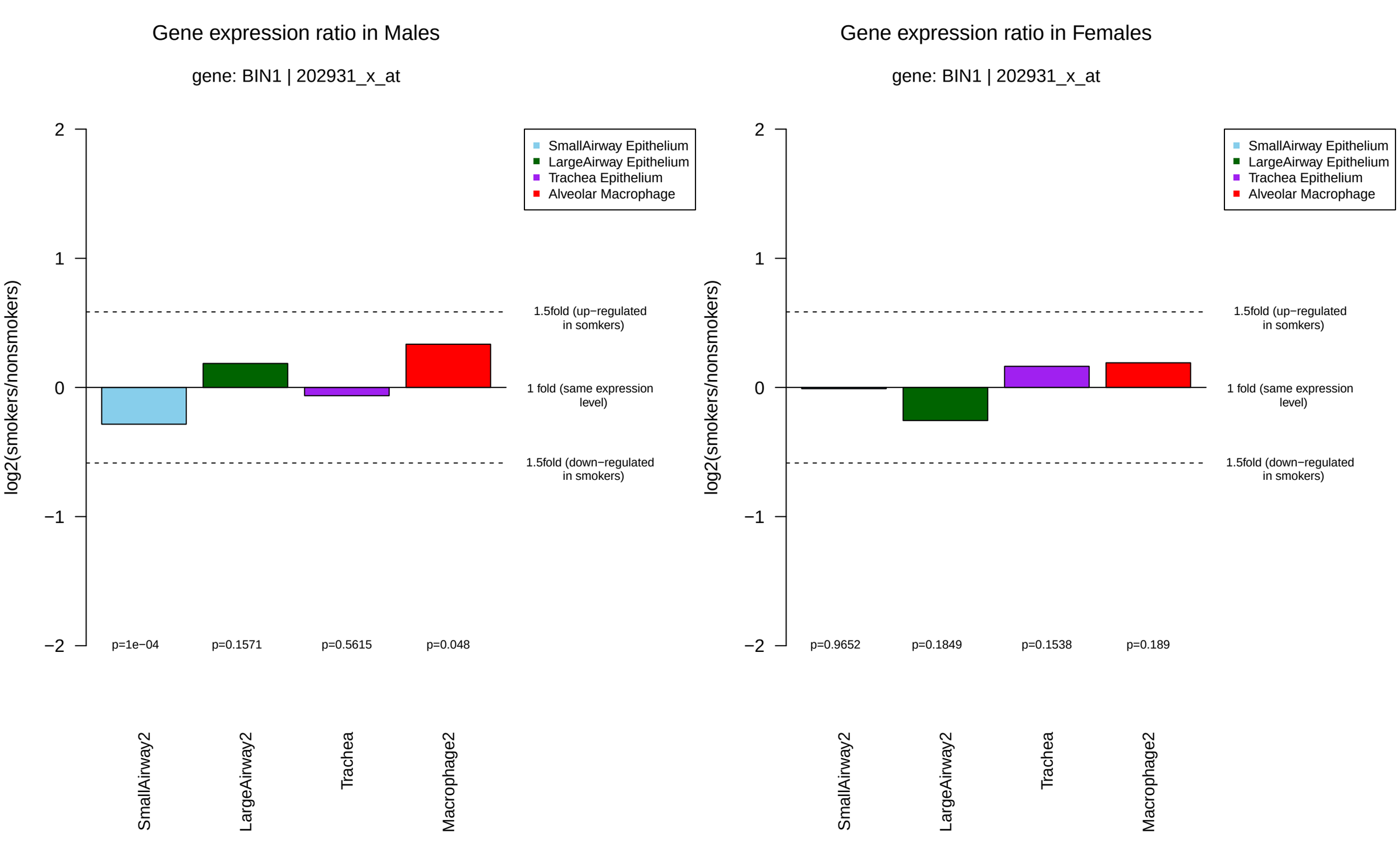
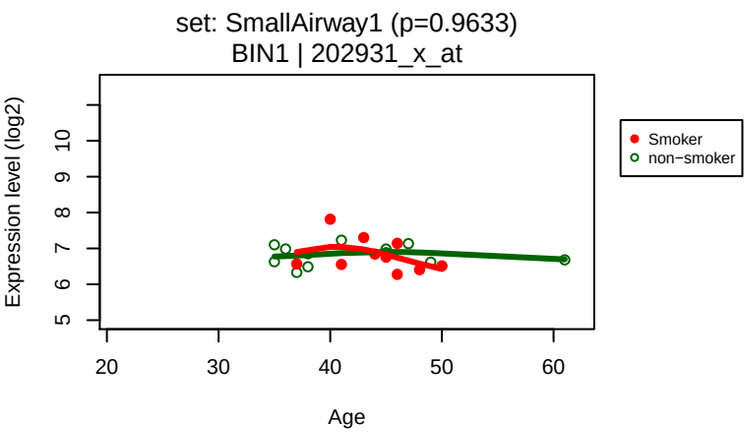
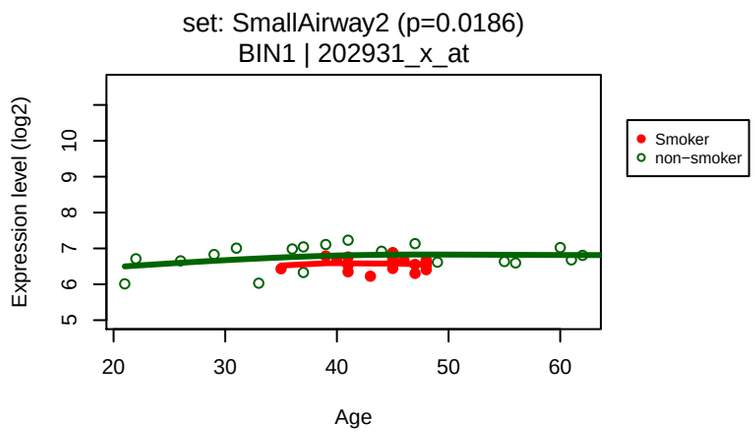
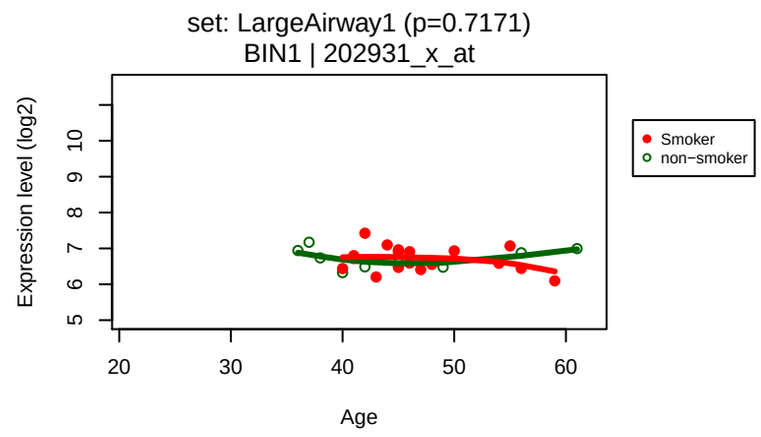

Gene expression of smokers and nonsmokers
gene: BIN1 | 202931_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9633
p=0.0186
p=0.0141
p=0.7171
p=0.7486
p=0.9491
p=0.1235
p=0.0175
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

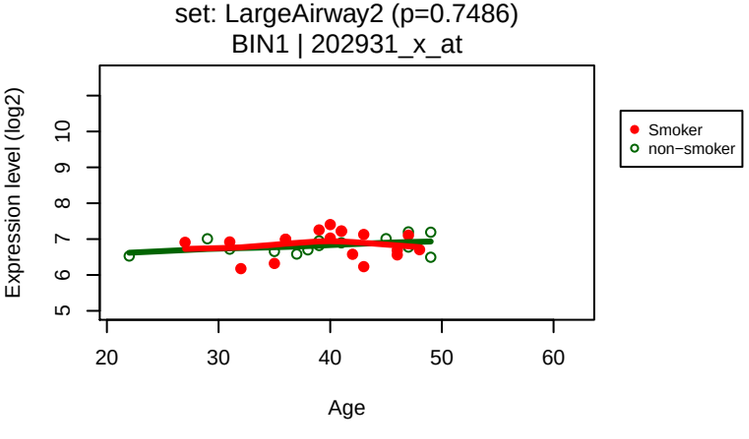

Gene expression ratio (smokers/nonsmokers)
gene: BIN1 | 202931_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9633
p=0.0186
p=0.0141
p=0.7171
p=0.7486
p=0.9491
p=0.1235
p=0.0175
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

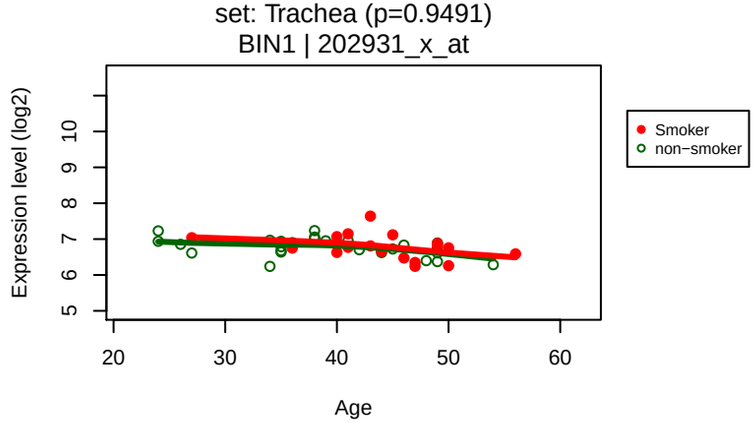

Gene expression ratio in Males
gene: BIN1 | 202931_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=1e−04
p=0.1571
p=0.5615
p=0.048
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BIN1 | 202931_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9652
p=0.1849
p=0.1538
p=0.189
SmallAirway2
LargeAirway2
Trachea
Macrophage2

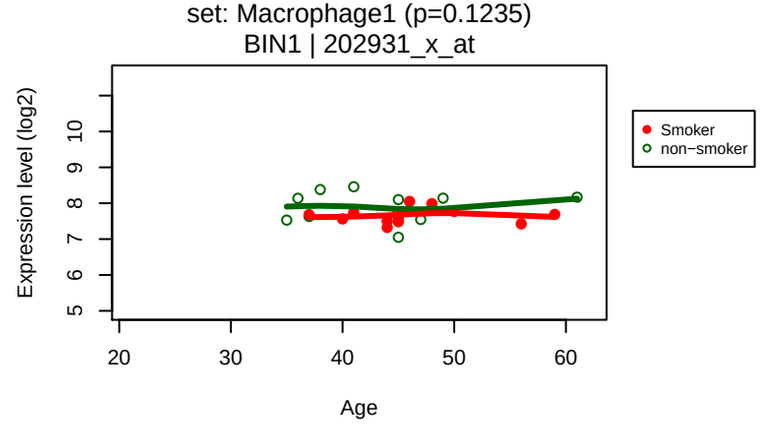

set: SmallAirway1 (p=0.9633)
BIN1 | 202931_x_at
Expression level (log2)
Age
Smoker
non−smoker

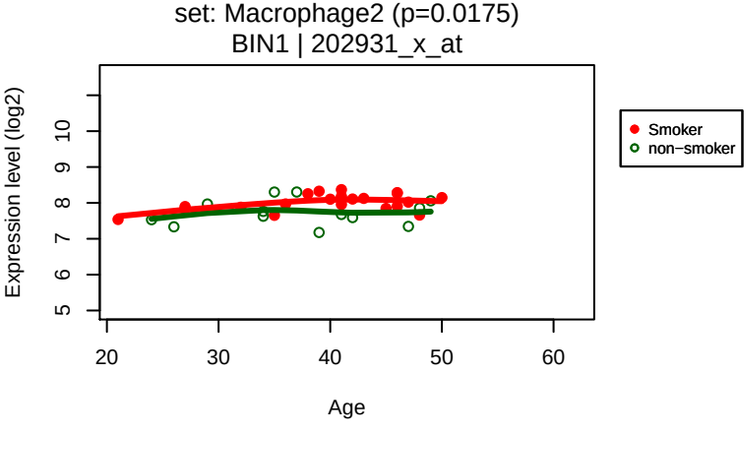

set: SmallAirway2 (p=0.0186)
BIN1 | 202931_x_at
Expression level (log2)
Age
Smoker
non−smoker

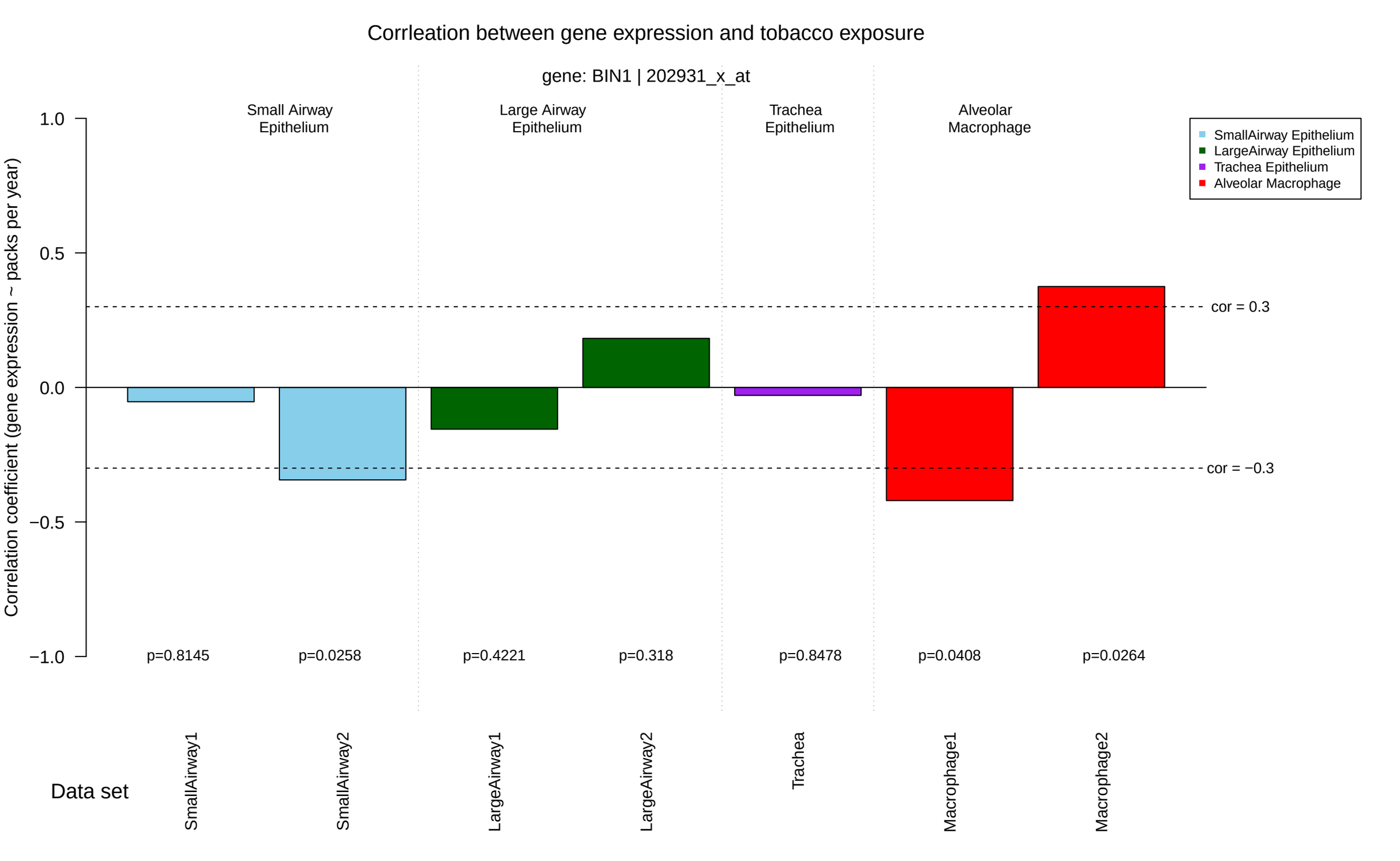

set: LargeAirway1 (p=0.7171)
BIN1 | 202931_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7486)
BIN1 | 202931_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9491)
BIN1 | 202931_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1235)
BIN1 | 202931_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0175)
BIN1 | 202931_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BIN1 | 202931_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8145
p=0.0258
p=0.4221
p=0.318
p=0.8478
p=0.0408
p=0.0264
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2